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ABSTRACT

Non-axisymmetric magnetic fields arising in a tokamak either by external or internal perturbations can induce complex non-ideal MHD responses in their resonant surfaces while remaining ideally evolved elsewhere. This layer response can be characterized in a linear regime by a single parameter called the inner-layer Δ , which enables outer-layer matching and the prediction of torque balance to non-linear island regimes. Here, we follow strictly one of the most comprehensive analytic treatments including two-fluid and drift MHD effects and keep the fidelity of the formulation by incorporating the numerical method based on the Riccati transformation when quantifying the inner-layer Δ . The proposed scheme reproduces not only the predicted responses in essentially all asymptotic regimes but also with continuous transitions as well as improved accuracies. In particular, the Δ variations across the inertial regimes with viscous or semi-collisional effects have been further resolved, in comparison with additional analytic solutions. The results imply greater shielding of the electromagnetic torque at the layer than what would be expected by earlier work when the viscous or semi-collisional effects can compete against the inertial effects, and also due to the intermediate regulation by kinetic Alfvén wave resonances as rotation slows down. These are important features that can alter the non-axisymmetric plasma responses including the field penetration by external fields or island seeding process in rotating tokamak plasmas.

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I. INTRODUCTION

A magnetic perturbation can induce complicated plasma responses in the neighborhood of the resonant surfaces where the wavefront of the perturbation is locally aligned with the main magnetic field. The plasma is no longer to be assumed ideally frozen to the magnetic field lines under the resonant perturbation but demands reconnecting of field lines,^{1,2} which could grow in time and bring a major modification in plasma profiles. Whether or not the reconnection region grows, initially, the resonant responses are bound only in spatially narrow layers compared to the outer layers where the plasma responses remain almost ideal. For this reason, the resonant response to a magnetic perturbation has been understood as a boundary layer phenomenon constrained by the matching conditions from outer layer responses. In particular, for a tokamak subjected to a magnetic perturbation, the initially linear inner-layer resonant responses can be completely characterized by a single parameter called the layer impedance or inner-layer Δ ,³ which is used to match the corresponding outer-layer solutions. The calculation of Δ is, indeed, central to understand and predict various tearing responses, which subsequently

determine the evolution or instability of tokamak plasmas. This is particularly true when the plasma temperature is sufficiently high, and, thus, resistivity is sufficiently low, since then, the linear layer width where non-ideal effects become important is greater than the width of resulting magnetic islands where non-linear effects become important.^{4,5}

One of such responses of great interest is the so-called field penetration.^{6,7} It refers to a bifurcation to large non-linear islands and reconnected fluxes expected due to the strong electromagnetic torque, which cannot simply be balanced in the linear framework. This is believed to be the main cause of the disruptive tokamak response called locked modes (LMs)^{8,9} under external magnetic perturbations, as the intrinsic (non-axisymmetric) error field problem in tokamaks^{10–19} is commonly called. The linear prediction would be no longer valid once the field deeply penetrates. However, the best strategy of error field control is to predict the onset of field penetration or the evolution in linear regimes before the onset, to entirely prevent LMs from taking place. It becomes increasingly difficult to control the LMs in a non-linear state although the control of LM itself has also been

successfully demonstrated.^{20,21} A closely related subject is the control of edge-localized modes (ELMs) using resonant magnetic perturbations (RMPs).^{22–28} The RMP subjects are different in many ways since it is intentional rather than intrinsic, and it induces transport across islands rather than avoiding it, in the edge rather than the core. Nonetheless, as implied in the literature,^{19,27,29–33} it entails initially the same dynamics of field penetration and, thus, follows very similar parametric dependencies. Another important area of applications would be on the onset of tearing modes in classical sense, which can play a role of seeding initial islands for neoclassical tearing modes (NTMs)^{34–36} before the neoclassically driven currents in the non-linear layer channel become important. The questions on the field penetration or classical tearing modes under linear layers are individually well defined but can also be solved together when Δ dependencies on the complex frequencies can be fully resolved.

Various MHD models have been used for many years to analytically predict the inner-layer Δ as well as its implications on the torque balance, field penetration either by error fields or RMPs, tearing mode instabilities, with their parametric dependencies. The resistive and inertial effects on Δ in stationary plasmas have been well addressed analytically, even in the full toroidal geometry.³⁷ However, other high order MHD effects including finite rotation, long mean-free-path physics such as semi-collisional or Hall effects, or drifts in two fluids have been successfully included only in a simplified geometry, e.g., a slab. The slab geometry around a sheared magnetic configuration is still justifiable in linear regimes since the layer remains narrow, and most of the geometric effects including the external boundary conditions come through outer layer responses. One of the highly successful, but still comprehensive, analytic model covering most of key linear regimes has been established by Fitzpatrick,^{38,39} Cole,^{40,41} and also Waelbroeck.^{29,42} These two-fluid drift-MHD models discussed in Refs. 38–41 are four-field models, extended from the original four-field model by Hazeltine *et al.*⁴³ to high- β plasmas conditions with additional phenomenological transport terms. Their linear results in Refs. 40 and 41 are, nonetheless, technically equivalent to the ones of three-field models in Refs. 29 and 42 if one ignores the coupling to the parallel ion motion. Note that there are also extended models, especially by Fitzpatrick and Waelbroeck,^{31,32,44–46} on drift MHD, which include important kinetic effects such as the neoclassical toroidal viscosity (NTV),^{12,47–49} which may also worth investigating in the linear framework.

This paper is organized as follows. Section II is devoted to revisiting the layer model in particular, by following Ref. 40, which is essential to clarify physics included in the presented results and later developments. Section III will introduce additional analytic examples, especially to understand and verify the response transitions across the inertial regimes found in the numerical simulations. The numerical schemes along with the Riccati transformation will be discussed in Sec. IV, and the primary results of the inner-layer Δ calculations and comparisons with essentially all asymptotic regimes in Ref. 40 will be presented in Sec. V, followed by concluding remarks in Sec. VI.

II. REDUCED TWO-FLUID DRIFT-MHD EQUATIONS FOR LAYERS—REVISITED

The layer model equation studied in this paper is the outcome of two-fluid drift-MHD formulation reduced in the slab geometry. The formulation is identical to the ones in the literature including Ref. 40, but still worth a presentation here for clarity. The adopted generalized

Ohm's law, the equation of motion, and the pressure evolution can be written as

$$\vec{E} + \vec{v} \times \vec{B} = \eta \vec{J} - \frac{1}{en_0} (\vec{\nabla} p_e - \vec{J} \times \vec{B}), \quad (1)$$

$$m_i n_0 \left(\frac{\partial}{\partial t} + \vec{v} \cdot \vec{\nabla} \right) \vec{v} = \vec{J} \times \vec{B} - \vec{\nabla} P + \mu_i \nabla^2 \vec{v}, \quad (2)$$

$$\left(\frac{\partial}{\partial t} + \vec{v} \cdot \vec{\nabla} \right) P = -\frac{5}{3} P (\vec{\nabla} \cdot \vec{v}) + \kappa \nabla^2 P, \quad (3)$$

where the fluid velocity $\vec{v} \cong \vec{v}_i = \vec{v}_E + \vec{v}_*$ with the $\vec{E} \times \vec{B}$ rotation $\vec{v}_E$ and the ion (electron) diamagnetic rotation $V_{i(e)*} = +(-)\hat{b} \times \vec{\nabla} p_{i(e)}/en_0 B$. The background density n_0 represents the density of both ions and electrons with the quasi-neutrality condition, i.e., $n_i = n_e = n_0$. The model obviously includes resistivity by η , the Hall effect in the last term in Eq. (1), phenomenological ion viscosity by μ_i , and thermal conductivity by κ .

Notice that the parallel ion motion $V_{||}$ is ignored here, which can be justified for a weak magnetic shear system such that $k_{||} c_s \ll \omega_{*i}$ within a thin resonant layer,²⁹ where c_s is the sound speed and ω_{*i} is the ion diamagnetic frequency. This assumption renders the final model equation, which will be shown later in Eq. (20), identical to the case where the parallel ion motion is included but with $(\vec{B} \cdot \vec{\nabla}) \vec{v} \approx 0$ as an approximation perhaps verifiable *a posteriori* depending on the parameters. This is because then the parallel motion of ions is decoupled from the evolution of the magnetic or the electric field, unless the electron viscosity is considered. In fact, the final model equation can be extended with electron viscosity if the same approximation is deployed as will be discussed in a separate paper. It is also interesting to see from Eq. (2) that only the $\vec{E} \times \vec{B}$ flow is convected. This is due to the cancellation between the convected diamagnetic flow and ion gyroviscosity, as is usually referred to as the gyroviscous cancellation.⁴³ Although the high-order gradient term of vorticity is ignored without a rigorous justification, the formulation above includes a good part of gyroviscous effects.

The governing equations can be reduced for a sheared-slab Cartesian geometry with a magnetic field given by $\vec{B} = \vec{\nabla} \psi \times \hat{z} + (B_0 + b_z) \hat{z}$, where B_0 is a constant magnetic field along with the symmetry direction $\hat{z}$, i.e., $\partial/\partial z = 0$. The radial direction is represented by $x = r - r_s$ with r is the radial flux label and $r = r_s$ is the radial position of the resonant surface having $q = m/n$. The other perpendicular direction to the equilibrium magnetic field is where the perturbation changes along with the wavenumber k , as represented by $y = (m\theta - n\phi)/k$, where θ and ϕ are the poloidal and toroidal angles, respectively. By normalizing the variables and parameters and by decoupling the compressional Alfvén wave as detailed in Ref. 39, one can develop a drift-MHD four-field model, which then becomes the three-field model without the parallel ion motion as

$$\frac{\partial \psi}{\partial t} = [\phi, \psi] - [Z, \psi] + \eta J, \quad (4)$$

$$\frac{\partial Z}{\partial t} = [\phi, Z] + d_\beta^2 [J, \psi] + (c_\beta^2 \eta + (1 - c_\beta^2) \kappa) Y, \quad (5)$$

$$\frac{\partial U}{\partial t} = [\phi, U] - \frac{\tau}{2} \{ \nabla^2 [\phi, Z] + [U, Z] + [Y, \phi] \}, \quad (6)$$

$$+ [J, \psi] + \mu_i \nabla^2 (U + \tau Y), \quad (7)$$

with the Poisson bracket $[A, B] = \hat{z} \cdot (\vec{\nabla} A \times \vec{\nabla} B)$. Note that all the variables and parameters in the aforementioned equations are normalized without units, which will need to be restored for the applications, as will be informed later for the final six parameters.

Three fields (ψ, Z, U) are representing the two components of magnetic fields and the $\vec{E} \times \vec{B}$ component of flow, with the helical flux ψ , $b_z = Z(1 + \tau)/d_i$, where $\tau = T_i/T_e$ and the ion skin-depth $d_i = \sqrt{m_i/n_0 e^2 \mu_0}/a$ with a convenient scale length a , $U = \nabla^2 \phi$, where $\vec{V}_E = \vec{\nabla} \phi \times \hat{z}$, and additionally with $Y = \nabla^2 Z$ and the current $J = \nabla^2 \psi$. Other normalized parameters are defined as $c_\beta = \sqrt{\beta/(1 + \beta)}$ with $\beta = (5/3)P_0/B_0^2$ with the equilibrium magnetic pressure P_0 and the toroidal magnetic field B_0 at the layer, and $d_\beta = c_\beta d_i/\sqrt{1 + \tau}$. The aforementioned three equations can be derived in a slab geometry by taking the $\hat{z} \cdot \vec{\nabla} \times$ operations to generalized Ohm's law, and the $\hat{z} \cdot \vec{\nabla} \times$ operation to the equation of motion, respectively, together with the radial force balance and pressure evolution under the ordering scheme of eliminating the compressional Alfvén wave.

The above derivations, in addition, assume that the density is uniform and constant and also that τ is uniform and constant, while the temperature is allowed to vary implicitly through the pressure variation. These assumptions are by and large for simplicity, limiting the application to the core field penetration, e.g., locked modes, where the density response seems to be much weaker than the temperature response.^{50–52} The model equation, nonetheless, remains almost identical in its final form, Eq. (20), even if the density variation is assumed as long as the temperature variation is ignored²⁹ or assumed as a function of the density variation alone.⁵³ This is because $b_z = -p_1 \propto -n_e T_e(1 + \tau)$, where p_1 is the leading-order variable pressure on the top of the uniform and constant P_0 in the ordering scheme that eliminates the compressional Alfvén wave,³⁹ enabling one to adopt one of (b_z, p_1, n_e) as an explicit variable albeit with different assumptions and interpretations. The model applicability is, therefore, extendable to the edge field penetration by RMPs where the density response appears to be predominant as is well known by the “pump-out” phenomena. A multiple-field model including the decoupled variations between density and temperature is difficult to be resolved analytically and has been treated successfully only in numerical simulations.^{27,54}

Now assuming a perturbation $\psi \propto e^{iky - i\omega t}$ to the background equilibrium $\psi_0(x) = -x^2/2$, $Z_0(x) = V_{e*}x$, and $\phi_0(x) = -Vx$, one can linearize the equations for the perturbations $(\psi, \tilde{Z}, \tilde{U})$. By introducing a stretch variable $x = \epsilon X$ to describe the narrow resonant layer, one can retain all the parameters by ordering rotation and ion skin depth on the order of $\sim \mathcal{O}(\epsilon)$, and resistivity, conductivity, and viscosity on the order of $\sim \mathcal{O}(\epsilon^3)$. That is, $V - \omega/k = \epsilon Q$, $V_{i(e)*} = -\epsilon Q_{i(e)}$, $d_\beta = \epsilon D$, $\eta = \epsilon^3 k$, $\kappa = \epsilon^3 kK$, and $\mu_i = \epsilon^3 kP$ with the renormalized variables.

The ordering scheme also produces the ideal MHD equations for the outer layer as desired. Without stretching

$$ix(\tilde{Z} - \tilde{\phi}) = i\epsilon(Q_e - Q)\tilde{\psi} + \epsilon^3 \nabla^2 \tilde{\psi}, \quad (8)$$

$$iQ\tilde{Z} - iQ_e\tilde{\phi} = i\epsilon D^2 x \nabla^2 \tilde{\psi} + \epsilon^2 C^2 \nabla^2 \tilde{Z}, \quad (9)$$

$$ix \nabla^2 \tilde{\psi} = i\epsilon(Q - Q_i) \nabla^2 \tilde{\phi} - \epsilon^3 P \nabla^4 (\tilde{\phi} + \tau \tilde{Z}), \quad (10)$$

where $C^2 \equiv (c_\beta^2 + (1 - c_\beta^2)K)$ and $\nabla^2 = (d^2/dx^2 - k^2)$. This shows in the leading order equilibrium that $\tilde{\phi} = \tilde{Z} = 0$ and $\nabla^2 \tilde{\psi} = 0$ for

outer layer, except the region near $x \sim 0$. The outer layer solution is, therefore, disconnected around the singular point $x = 0$, producing the jump in the derivative,

$$\Delta \equiv \left. \frac{d \ln \tilde{\psi}}{dx} \right|_{-}^{+} = \Delta' + \Delta'_{EF}. \quad (11)$$

As well known, Δ' and Δ'_{EF} denote the internal drive due to the tearing mode and the external drive due to error fields, respectively. In practice, these outer-layer quantities would better be calculated in realistic toroidal geometry rather than the slab. Unlike the thin inner layers, the poloidal mode coupling in most of tokamaks is not ignorable in the outer layers. For example, Δ'_{21} at the $q = 2/1$ rational surface is not determined only by the $(m = 2, n = 1)$ component of perturbations but by many other poloidal m components even in a linear regime.⁵⁵ The purpose of this paper is to develop a reliable method to solve the inner layer equations and to obtain the parametric dependencies of Δ . Nonetheless, an interesting future work is to integrate the inner-layer Δ calculations in this paper into the readily available outer-layer solvers such as resistive DCON or IPEC,^{55,56} which can provide realistic Δ' or Δ'_{EF} .

Now stretching the variable for the layer

$$i(Q - Q_e)\tilde{\psi} = iX(\tilde{\phi} - \tilde{Z}) + \frac{d^2 \tilde{\psi}}{dX^2} + \mathcal{O}(\epsilon^2), \quad (12)$$

$$iQ\tilde{Z} - iQ_e\tilde{\phi} = iD^2 X \frac{d^2 \tilde{\psi}}{dX^2} + C^2 \frac{d^2 \tilde{Z}}{dX^2} + \mathcal{O}(\epsilon^2), \quad (13)$$

$$i(Q - Q_i) \frac{d^2 \tilde{\phi}}{dX^2} = iX \frac{d^2 \tilde{\psi}}{dX^2} + P \frac{d^4 (\tilde{\phi} + \tau \tilde{Z})}{dX^4} + \mathcal{O}(\epsilon^2). \quad (14)$$

Note that the only simplification deployed here compared to the full drift MHD equations in the slab, i.e., Eqs. (4)–(7), is $\nabla^2 \sim d^2/dX^2$. Applying the generalized Fourier–Laplace transformation, i.e., $\tilde{\phi}(p) = \int_c \tilde{\phi}(X) e^{ipX} dX$ with the appropriate contour path c in the complex- X plane,⁴² one can obtain

$$i(Q - Q_e)\tilde{\psi} = \frac{d(\tilde{\phi} - \tilde{Z})}{dp} - p^2 \tilde{\psi}, \quad (15)$$

$$iQ\tilde{Z} - iQ_e\tilde{\phi} = -D^2 \frac{d(p^2 \tilde{\psi})}{dp} - C^2 p^2 \tilde{Z}, \quad (16)$$

$$i(Q - Q_i)p^2 \tilde{\phi} = \frac{d(p^2 \tilde{\psi})}{dp} - P p^4 (\tilde{\phi} + \tau \tilde{Z}). \quad (17)$$

Rearranging the above equations, it becomes convenient to define

$$Y \equiv \tilde{\phi} - \tilde{Z} = F(p)\tilde{\phi}, \quad (18)$$

giving

$$\tilde{\psi} = \frac{1}{i(Q - Q_e) + p^2} \frac{dY}{dp}, \quad (19)$$

$$\frac{d}{dp} \left[\frac{p^2}{i(Q - Q_e) + p^2} \frac{dY}{dp} \right] - p^2 G(p)Y = 0, \quad (20)$$

with

$$F(p) \equiv \frac{i(Q - Q_e) + (C^2 + i(Q - Q_i)D^2)p^2 + (1 + \tau)PD^2p^4}{iQ + C^2p^2 + \tau PD^2p^4}, \quad (21)$$

$$G(p) \equiv \frac{-Q(Q - Q_i) + i(Q - Q_i)(P + C^2)p^2 + C^2Pp^4}{i(Q - Q_e) + (C^2 + i(Q - Q_i)D^2)p^2 + (1 + \tau)PD^2p^4}, \quad (22)$$

again as derived in Ref. 40. It has already been shown that the boundary conditions for $Y(p)$ are $\lim_{p \rightarrow +\infty} Y(p) = 0$ (or bounded) and $\lim_{p \rightarrow +0} Y(p) = Y_0(1 + \hat{\Delta}/(\pi p)) + \mathcal{O}(p)$ from the small- p expansion, where $\hat{\Delta} = \epsilon\Delta$.

Note again that this final equation is essentially identical to the equation derivable with the variable density in other models.^{29,53} The only main difference in those models is that the resistive diffusion represented here by $C^2 = (c_\beta^2 + (1 - c_\beta^2)K)$ is replaced by the cross field particle diffusion, $D_\perp$, in a normalized form. In particular, the latest Ref. 53 emphasized its important consequence since $D_\perp$ is not as small as C^2 in the realistic tokamak conditions. The future application of the formulation in this paper to the new diffusive regimes will be straightforward by assuming $C^2 \sim \mathcal{O}(1)$ or $C^2 \gg 1$.

The parameter set assigned to solve Eq. (20) and, thus, to evaluate $\hat{\Delta}$ is related to the tokamak parameters as follows: define the characteristic hydromagnetic, resistive, and viscous time scales as $\tau_H = R_0(\mu_0\rho(r_s))^{1/2}/ns(r_s)B_0$, $\tau_R = \mu_0 r_s^2/\eta(r_s)$, and $\tau_V = r_s^2\rho(r_s)/\mu_i(r_s)$, respectively, where r_s is the minor radius of the resonant layer, R_0 is the major radius of the magnetic axis, $\rho(r_s)$ is the mass density at the layer, $s(r_s) = (1/q(r_s))(dq/dr)_{r_s}$ is the magnetic shear at the layer, and $q(r)$ is the safety factor profile along the minor radius. The Lundquist number is then given by $S = \epsilon^{-1/3} = \tau_R/\tau_H$. Let $\vec{E} \times \vec{B}$ be the angular frequency ω_E and the diamagnetic angular frequencies $\omega_{i(e)*}$. Then, the parameter set (Q, Q_e, Q_i, C, D, P) is estimated by $Q = S^{1/3}\omega_E\tau_H$ ignoring the mode rotation, i.e., $\omega = 0$, $Q_{i(e)} = -S^{1/3}\omega_{i(e)*}\tau_H$, $C = c_\beta$ ignoring the thermal conductivity for now, $D = S^{1/3}\rho_s/r_s$ where ρ_s is the ion Larmor radius based on the electron temperature, and $P = \tau_R/\tau_V$.

Equation (20) is still not analytically tractable by itself but convenient enough to systematically study various parametric regimes. The previous studies have shown more than ten characteristically different regimes, including the first and second Visco-Resistive regimes (VRi, VRii), Hall-Resistive regimes (HRi, HRii), Semi-Collisional regimes (SCi, SCii), Resistive-Inertial regimes (Rli, Rlii), Visco-Inertial (VI), and Inertial regime (I). Each regime is essentially quantified with $\hat{\Delta}$, by keeping only one term out of the rational function $G(p)$ depending on relative importance among the parameters (Q, Q_e, Q_i, C, D, P) . Each of $\hat{\Delta}$ can then be used to predict the electromagnetic torque variation across the layer due to magnetic perturbations as a function of rotation by⁴⁰

$$\tau_\phi \propto -\frac{\Im(\hat{\Delta}(Q))}{|\hat{\Delta}' + \hat{\Delta}(Q)|} \approx \Im(1/\hat{\Delta}(Q)), \quad (23)$$

ignoring the contribution from the conventional tearing mode index $\hat{\Delta} = \epsilon\Delta'$. Note that the same normalization $\hat{\Delta} = \epsilon\Delta$ and so our assumption is not justified by $\epsilon \ll 1$ but just $\Delta' \ll \Delta$, which is taken for simplicity. It will be necessary to estimate Δ' in practical cases to find its relative importance and must be restored in the calculations of torque balance as needed. This relation, Eq. (23), provides physical interpretation of $\hat{\Delta}$ in the rest of this paper. Generally, the real part turns out to be subdominant under the imaginary part of $\hat{\Delta}$ and so $\tau_\phi \propto 1/\Im(\hat{\Delta})$. That is, one can take $\Im(\hat{\Delta})$ as an approximate inverse of the resonant layer torque.

III. TRANSITIONS ACROSS INERTIAL REGIMES

The asymptotic $\hat{\Delta}(Q)$ calculations, however, remained unsatisfactory in the regimes where inertial effects become important, particularly in describing the field penetration process. This is because $\hat{\Delta}(Q) \propto 1/Q^\alpha$, where $\alpha > 0$ for the two inertial VI and I regimes, meaning that the electromagnetic torque becomes smaller when rotation slows down and can always be balanced by viscous torque without a penetration. On the other hand, the two inertial regimes are most relevant for the core, definitely for Neutral Beam Injected (NBI) plasmas in currently operating tokamak experiments and also likely for ITER. For example, $Q > 20 \gg D > 1$ for DIII-D high- β conditions, and $Q > 5 \gg D \sim 1$ for ITER $Q=10$ baseline scenario even with a very low torque assumption. It is then a transition from the inertial regimes to another regime where $\hat{\Delta}(Q)$ hits the maxima, or equivalently where the electromagnetic torque hits the minima as rotation slows down, that dominates the determination of the final torque balance. This section will show the two analytic examples.

First suppose high viscous and inertial regime, i.e., $Q, P \gg 1$ but $C, D \ll 1$. Equation (20) can be reduced to

$$\frac{d^2 Z}{dp^2} - i(Q - Q_i)(iQ + Pp^2)Z \simeq 0, \quad (24)$$

with $Z = pY$. This equation can then be transformed to a parabolic cylinder function with the solution

$$Z = D_\nu(\gamma p), \quad (25)$$

where

$$\gamma \equiv (4i(Q - Q_i)P)^{1/4}, \quad (26)$$

$$\nu \equiv -\frac{1}{2} + \frac{1}{2} \left[\frac{Q^2(Q - Q_i)}{iP} \right]^{1/2}. \quad (27)$$

Using the known results of the parabolic cylinder function, $\lim_{p \rightarrow 0} D_\nu \sim \pi^{1/2} 2^{\nu/2} / \Gamma(1/2 - \nu/2)$ and $\lim_{p \rightarrow 0} dD_\nu/dp \sim -\pi^{1/2} 2^{(\nu+1)/2} / \Gamma(-\nu/2)$, one can obtain

$$\hat{\Delta}_{VI-I} = -\frac{\pi}{2^{1/2}\gamma} \frac{\Gamma(-\frac{\nu}{2})}{\Gamma(\frac{1}{2} - \frac{\nu}{2})}, \quad (28)$$

which is the combined result of the VI and I regimes. When $P \gg Q^3$, $\nu \rightarrow -\frac{1}{2}$ and so

$$\hat{\Delta}_{VI} \sim -\frac{\pi \Gamma(1/4)}{2 \Gamma(3/4)} (i(Q - Q_i))^{-1/4} P^{-1/4}, \quad (29)$$

reproducing the $\hat{\Delta}$ in the Visco-Inertial regime. For the opposite limit $P \ll Q^3$, $|\nu| \rightarrow \infty$. Using $\Gamma(z+b)/\Gamma(z+a) \sim z^{a-b}$ as $|z| \rightarrow \infty$

$$\hat{\Delta}_I \sim -\frac{\pi}{2^{1/2}\gamma} \left(-\frac{\nu}{2}\right)^{1/2} \sim -\frac{\pi}{(iQ)^{1/2}(i(Q - Q_i))^{1/2}}. \quad (30)$$

This confirms that the formulation of the Inertial regime is correct as it is the analytic continuation in the parameter as $P \rightarrow 0$, whereas the solution strictly with $P=0$ from Eq. (24) is purely oscillatory as $p \rightarrow \infty$.

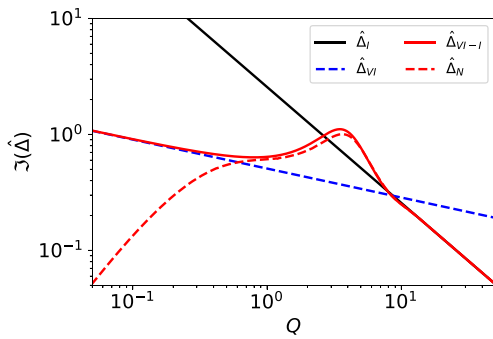


FIG. 1. The transition across the Visco-Inertial (VI) and Inertial Regime (I) analytically predicted ($\hat{\Delta}_{VI-I}$, red solid line), compared to the numerical solution ($\hat{\Delta}_N$, red dashed line) as well as the asymptotic behavior of each regime ($\hat{\Delta}_{VI}$, blue dashed line) and ($\hat{\Delta}_I$, black solid line).

Figure 1 shows the variations $\hat{\Delta}_{VI-I}(Q)$ in comparison with the $\hat{\Delta}_{VI}(Q)$ and $\hat{\Delta}_I(Q)$ when $P=100$ and $Q_e = Q/2$ and $Q_i = -Q/2$. The connection formula is revealing the peak in the transitions between the two regimes, implying the substantial reduction of the electromagnetic torque when the viscosity takes its effects fully through the inertial layer, i.e., when $Q \sim P^{1/2}$. Figure 1 also shows the comparison with the full numerical solution for the case with $C=0.1$, $D=0.2$, and $\tau=1.0$ in addition to the same P , verifying the validity of the analytic transition when P is high enough. The difference becomes important when $P < 10$, indicating the need of the numerical approach for precise quantifications as will be described in Sec. IV.

It is also interesting to investigate the analytic structure under the transition to the inertial regime when $D \gg 1$. Assuming $C, P \ll 1$ for this limit, the Eq. (20) can be reduced to

$$\hat{\Delta}_{S-I} = -i \frac{\pi \delta}{F_2(\nu) \tan^2\left(\frac{\pi \nu}{2}\right)} \left[\frac{1 + i \tan\left(\frac{\pi \nu}{2}\right) - \left(\frac{\delta}{2}\right)^{2\nu+1} F_1(\nu) d_0 \left(2 \tan\left(\frac{\pi \nu}{2}\right) - \tan(\pi \nu) \left(1 + i \tan\left(\frac{\pi \nu}{2}\right)\right)\right)}{1 - i \cot\left(\frac{\pi \nu}{2}\right) + \left(\frac{\delta}{2}\right)^{2\nu+1} F_1(\nu) d_0 \left(2 \cot\left(\frac{\pi \nu}{2}\right) + \tan(\pi \nu) \left(1 - i \cot\left(\frac{\pi \nu}{2}\right)\right)\right)} \right], \quad (36)$$

with

$$F_1(\nu) = \frac{\Gamma(\nu) \Gamma(\nu+2)}{\Gamma\left(\nu + \frac{1}{2}\right) \Gamma\left(\nu + \frac{3}{2}\right)}, \quad (37)$$

$$F_2(\nu) = \frac{(\nu+1) \Gamma\left(1 - \frac{\nu}{2}\right) \Gamma\left(\frac{1}{2} + \frac{\nu}{2}\right)}{\Gamma\left(1 + \frac{\nu}{2}\right) \Gamma\left(\frac{1}{2} - \frac{\nu}{2}\right)}. \quad (38)$$

Note that the result above is different from a similar calculation in Ref. 58. It appears that the authors in the reference considered only a leading order term from each P_ν and Q_ν , while their full next order corrections are required to calculate the $\hat{\Delta}_{S-I}$. In the limit of $Q \gg 1$,

$$\frac{d^2 Z}{dp^2} + \frac{i(Q - Q_e)Q(Q - Q_i)}{i(Q - Q_e) + (C^2 + i(Q - Q_i)D^2)p^2} Z \simeq 0. \quad (31)$$

This equation has the general form of the solution

$$Z = (1 + p^2/\delta^2) [c_1 P'_\nu(ip/\delta) + c_2 Q'_\nu(ip/\delta)], \quad (32)$$

with the Legendre functions of the first and second kind, where the prime denotes the differentiation with respect to the argument and

$$\delta \equiv \left[\frac{i(Q - Q_e)}{C^2 + i(Q - Q_i)D^2} \right]^{\frac{1}{2}}, \quad (33)$$

$$\nu \equiv -\frac{1}{2} + \left[\frac{1}{4} - \frac{i(Q - Q_e)Q(Q - Q_i)}{C^2 + i(Q - Q_i)D^2} \right]^{\frac{1}{2}}. \quad (34)$$

Note here that C is kept even if it is small, as otherwise, it can make $\Re(\nu) = -\frac{1}{2}$ and make the asymptotic limits degenerate. The asymptotic behavior for the large p of $Y = Z/p$ can then be represented by

$$Y \sim (ip)^{-\nu} + d_0(ip)^{1+\nu} \text{asp} \rightarrow \infty. \quad (35)$$

Here, d_0 is representing the contributions from the dominant power-like solution, as a function of the original ratio c_1/c_2 and the parameters δ and ν , but unfortunately remains undetermined within this formulation. The solution in Eq. (32) becomes intermediate due to the finite D , requiring the larger p to ensure its connection to the exponentially small solution where the other ignored terms come into play. Nonetheless, it is the purpose of this investigation to understand the underlying structure, possibly by patching d_0 to reproduce the result in a neighboring regime.

Using the small- p expansion of the Legendre functions,⁵⁷ the following result is obtained for the $\hat{\Delta}$:

interestingly, but somewhat expectedly, it reproduces the inertial regime regardless of d_0 . This can be seen since then $\nu \rightarrow i\infty$, $\tan\left(\frac{\pi \nu}{2}\right) \rightarrow i$, making $[\dots] \rightarrow 1$ in Eq. (36), and with $F_2 \rightarrow i\nu$,

$$\Delta_{S-I} \sim \frac{\pi \delta}{\nu} \sim \Delta_I. \quad (39)$$

Now although it is not asymptotically correct, one can use the freedom in d_0 to patch the result to another limiting behavior in the neighboring regime, the second Semi-Collisional regime, as $Q \rightarrow 0$ and $\nu \rightarrow 0$. So, we have

$$\Delta_{S-I} \sim i \frac{\pi}{d_0} \approx \Delta_{S-Cii} = \frac{\pi(iQ)^{1/2}i(Q - Q_e)}{D}, \quad (40)$$

leading to choose $d_0 = D/((iQ)^{1/2}(Q - Q_i))$ for the connection between the Semi-Collisional and Inertial regimes. An example is

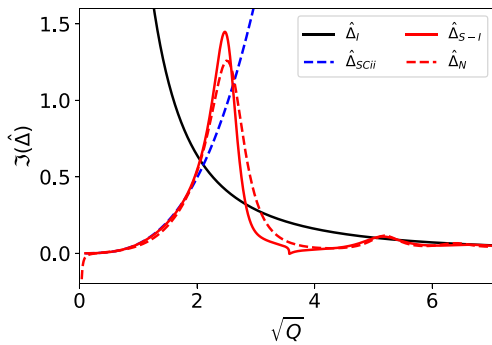


FIG. 2. The transition across the second Semi-Collisional (SCii) and Inertial Regime (I) analytically predicted ($\hat{\Delta}_{S-I}$, red solid line), compared to the numerical solution ($\hat{\Delta}_N$, red dashed line) as well as the asymptotic behavior of each regime ($\hat{\Delta}_{SCII}$, blue dashed line) and ($\hat{\Delta}_I$, black solid line). Note that the x-axis scale is adjusted with $\sqrt{Q}$ to better represent the structure.

shown in Fig. 2 in the comparisons with the $\hat{\Delta}_{SCII}(Q)$ and $\hat{\Delta}_I(Q)$ when $D = 15$ and $Q_e = -Q_i = Q/2$ as well as the numerical result with $C, P \ll 1$. The sharp change seen in $\hat{\Delta}_{S-I}$ (near $\sqrt{Q} \sim 4$) is due to the transition of $\nu(Q)$ from a real to complex value near $\nu \sim -1/2$ as Q increases.

There are two meaningful aspects shown from this exercise, which are (1) again the greater peak values of $\hat{\Delta}$, or shielding of electromagnetic torque, than the simple connection during the transition between the SCii and I regimes, and (2) the existence of oscillatory response when rotation is large enough but before entering the full Inertial regime (i.e., $Q \rightarrow \infty$). This oscillatory response, in fact, has been identified as the Kinetic Alfvén Wave (KAW) regime in the context of resonant layer response in other previous studies.⁵⁹ The oscillatory behavior is associated with $(\delta/2)^{2\nu}$ in Eq. (36). As $\delta \sim D^{-1}$ and $\nu \sim -\frac{1}{2} + iQ/D$ if $Q > D$, one can see the characteristic frequency of the oscillation along with Q is $f_Q \sim (2\pi/\ln(2D))D$, almost proportionally to D . By removing the normalization, this means $f_V \sim V_A d_B \sim V_A(\rho_s/a)$ if $\beta \ll 1$. Here, V_A is the Alfvén velocity, and ρ_s is the ion Larmor radius calculated using the electron temperature, indicating the relevance of the KAW identification. The KAW response is expected to disappear, exponentially along with Q as $\nu \rightarrow +i\infty$, $\tan(\pi\nu) \rightarrow i$, or $\tan(\frac{\pi\nu}{2}) \rightarrow i$, and, thus, $[\dots] \rightarrow 1$ in Eq. (36) exponentially. These are all qualitatively consistent with the numerical solutions as will be shown later. However, in order to quantitatively reproduce the transition and oscillatory behavior due to the sizable D by an analytic method, one will need to find the large- p layer that could asymptotically match the form in Eq. (35), which has not been formulated yet. So numerically quantified solutions will be used to address a potential physical impact of the transient KAWs through Secs. IV–VI.

IV. RICCATI TRANSFORMATION FOR LAYER EQUATIONS

The issue in the analytic approaches after deriving the single layer Eq. (20) is that there is always a range of $p \in (0, \infty)$ where any of the single terms, either in the numerator or denominator of the rational functions, can become dominant. Switching to a numerical approach at this point is, therefore, important to keep the quantitative applicability of the drift MHD slab model. On the other hand, a numerical

calculation of (20) can also be subtle due to the infinite boundary, and the conditions defined at the singular points. The $p = 0$ is a regular singular point and $p = \infty$ is an irregular singular point,⁶⁰ leaving the great possibilities to have the solution highly sensitive to the initial numerical condition and across widely different parameters. This issue can be greatly alleviated, by the Riccati transformation with $W = (p/Y)(dY/dp)$,⁶¹ since this essentially turns exponential steepness to algebraic behavior. The Riccati transformation yields

$$\frac{dW}{dp} = \left[\frac{2p}{i(Q - Q_e) + p^2} - \frac{1}{p} \right] W - \frac{W^2}{p} + p[i(Q - Q_e) + p^2]G(p). \quad (41)$$

The solution asymptotes to

$$W \sim \alpha p^2 \left(\frac{Ae^{\alpha p^2/2} - e^{-\alpha p^2/2}}{Ae^{\alpha p^2/2} + e^{-\alpha p^2/2}} \right) (p \rightarrow +\infty), \quad (42)$$

where $\alpha = C/(\sqrt{1 + \tau}D)$. The constant of integration A represents the ratio of the exponentially growing solution to the exponentially decreasing solution of Y . When integrating inversely from a large p to the left until approaching to $p \rightarrow 0$, the best initial condition $W_0(p_0)$ to choose is, therefore, $A = 0$ and so $W_0 = -\alpha p_0^2$, and the worst initial condition would be $A \rightarrow \infty$ and so $W_0 = +\alpha p_0^2$. With an arbitrary choice for $W_0(p_0)$, the ratio of the exponentially large to small solution is given by

$$A = \left(\frac{1 + \frac{W_0}{\alpha p_0^2}}{1 - \frac{W_0}{\alpha p_0^2}} \right) e^{-\alpha p_0^2}. \quad (43)$$

This shows that $A \ll 1$ as long as p_0 is large enough, so that $\alpha p_0^2 \gg 1$. On the contrary, the power series expansion for $p \rightarrow 0$ yields

$$W \sim -1, \quad \frac{dW}{dp} \sim \frac{\pi}{\Delta} (p \rightarrow +0). \quad (44)$$

That is, the exponentially large $Y(p)$ in $p \rightarrow \infty$ which is physically irrelevant is naturally suppressed by choosing the initial p_0 large enough in the backward integration of $W(p)$, and the inverse of the asymptotic gradient of $W(p)$ as $p \rightarrow +0$ will give the $\hat{\Delta}$.

It becomes then straightforward to implement the Riccati Eq. (41) and numerically calculate $\hat{\Delta}(Q, Q_e, Q_i, C, D, P)$. A new code named Slab Layer (SLAYER) has been developed based on the adaptive backward integration using the LSODE package. The default initial value is $W_0 = -\alpha p_0^2$ as discussed, but it is found that the solution converges correctly even with the worst initial value $W_0 = +\alpha p_0^2$ as long as p_0 is sufficiently large, indicating the robustness of the Riccati solution. Figure 3 shows four different initial values of p_0 in the case when $W_0 = -\alpha p_0^2$, and two additional cases with $p_0 = 10$ but with $W_0 = 0$ and $W_0 = +\alpha p_0^2$. All cases show $W \rightarrow -1$ as expected, and a good convergence to the same value of dW/dp , i.e., $\hat{\Delta}$ except when $p_0 = 2$. It is interesting to see that the worst assignment $W_0 = +\alpha p_0^2$ exhibits a pole-like behavior but still converges when p_0 is sufficiently large.

This robust behavior is also seen by another test, Fig. 4, showing that the values of $\hat{\Delta}$ remain reliable even when $W_0(\theta) = \alpha p_0^2 e^{i\theta}$, i.e.,

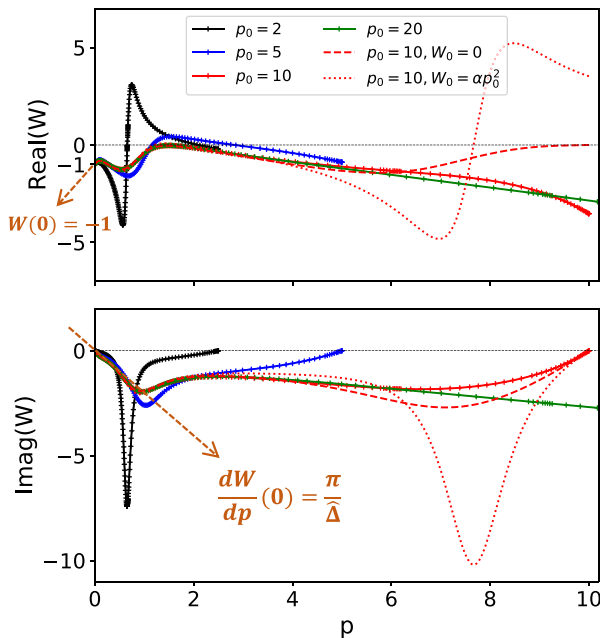


FIG. 3. Convergence of the Riccati solutions for W as $p \rightarrow +0$ with various p_0 starting points, $p_0 = 2$ (black solid line), $p_0 = 5$ (blue solid line), $p_0 = 10$ (red solid line), $p_0 = 20$ (green solid line) with $W_0 = -\alpha p_0^2$ for each, and also with two different initial W_0 values, $W_0 = 0$ (red dashed line) and $W_0 = \alpha p_0^2$ in the case of $p_0 = 10$. $W \rightarrow -1$ as $p \rightarrow +0$ expected with all (p_0, W_0) values and dW/dp also converges well except when $p_0 = 2$, which is not large enough. Here, the six parameters (Q, Q_e, Q_i, C, D , and P) are (4.9, 0.98, -0.98, 0.082, 4.36, and 0.50, respectively), which correspond to the conditions such as $T_{e,i} = 1$ keV, $n_0 = 10^{19} \text{ m}^{-3}$, and $\omega = 100 \text{ krad/s}$, locally at $q = 2/1$ with the magnetic shear $s = 2.0$ for a mid-size tokamak with $a = 0.5 \text{ m}$, $R = 1.0 \text{ m}$, and $B_0 = 1 \text{ T}$.

assigned by complex values, when p_0 is sufficiently large. One can see that the initial conditions near the best recommended choice (near the white dash line $\theta \sim \pi$ or $W_0 \sim -\alpha p_0^2$) give the gradual convergence to the answer (blue on the right) when p_0 increases in $p_0 < 10$.

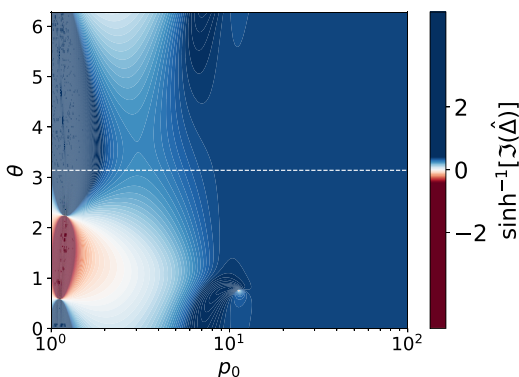


FIG. 4. Convergence of the Riccati solutions for W as $p \rightarrow +0$ shown as a function of (p_0, θ) , with the complex value $W_0 = \alpha p_0^2 e^{i\theta}$. The white dash line indicates the best recommended choice $W_0 = -\alpha p_0^2$. Note that the contour has the $\sinh^{-1}$ scale, which is similar to a log scale but can include the sign change. The six parameters (Q, Q_e, Q_i, C, D , and P) are same as Fig. 3.

This convergence is better and faster than the choices near the other end ($\theta = 0, 2\pi$), where the answers vary unexpectedly with local and random extrema, as shown in the contour, before the final convergence in $p_0 > 10$. Nonetheless, the convergence is eventually obtained with any initial conditions when $p_0 > 20$, indicating the reliability of this numerical procedure. Of course, these convergent characteristics depend on the parameters (Q, Q_e, Q_i, C, D , and P). It is found that in some extreme conditions such as very high Q and low C, D , or P , the process can be numerically unstable if p_0 is too large. Those extreme conditions are not typical in realistic tokamaks but could occur in astrophysical conditions where extra cautions must be taken to ensure the convergence of the $\hat{\Delta}$ calculation.

V. NUMERICAL VERIFICATIONS FOR PARAMETRIC DEPENDENCIES

Riccati solutions were then compared with the asymptotic analytic solutions in the large (Q, P) space. For the purpose of the benchmark with analytic solutions, the electron and ion diamagnetic rotation are scaled linearly with $\vec{E} \times \vec{B}$, as $Q_e = Q/2$ and $Q_i = -Q/2$. It is perhaps realistic to fix the Q_e and Q_i , which can, however, complicate our comparisons due to the resonances nearby $Q \sim Q_{e,i}$. Other parameters are fixed, first with the values relevant for low- β tokamak conditions, with $C = 0.1$, $D = 0.2$, and simply $\tau = 1.0$. These conditions can be used to reproduce Fig. 1 in Ref. 40, but additionally with explicit $\hat{\Delta}$ values, as shown in Figs. 5(a) and 6(a), in comparison with the numerical solutions for $\hat{\Delta}$ in Figs. 5(b) and 6(b). There are eight different regimes identified in the figures, including the Inertial (I), Visco-Inertial (VI), Semi-Collisional (SCi), Visco-Resistive (VRi), Resistive-Inertial (RIi), Hall-Resistive (HRI), and also the second Visco-Resistive (VRii) and second Resistive-Inertial (RIii)

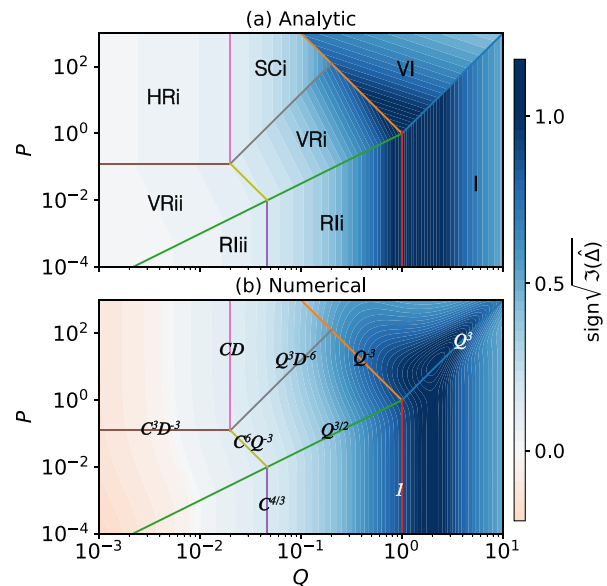


FIG. 5. Comparisons of (a) analytic and (b) numerical solutions for $\Im(\hat{\Delta})$ in the (Q, P) space with $C = 0.1$, $D = 0.2$, $\tau = 1$, and $Q_e = Q/2$ and $Q_i = -Q/2$. Note that the contour is on a square-root scale with the sign multiplied, to show the variations better than a linear scale. This scale is used for Fig. 6–8 as well. The regime identifications are equivalent to Fig. 1 in Ref. 40.

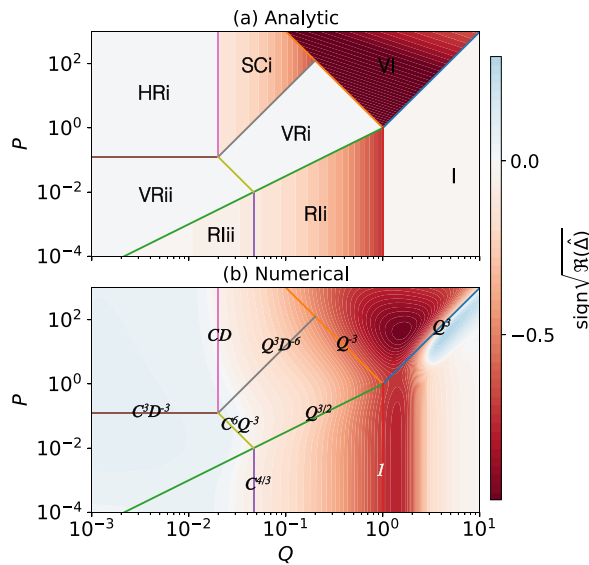


FIG. 6. Comparisons of (a) analytic and (b) numerical solutions for $\Re(\hat{\Delta})$ in the (Q, P) space with $C=0.1$, $D=0.2$, $\tau=1$, and $Q_e = Q/2$ and $Q_i = -Q/2$. The regime identifications are equivalent to Fig. 1 in Ref. 40.

regimes. Note that this diagram requires $C < D \ll C^{1/3}$, limiting the choice of C and D .

The comparison for $\Im(\hat{\Delta})$ in Fig. 5 shows that the analytic results represent the values reasonably unless $\Im(\hat{\Delta})$ becomes very small. This is somewhat more satisfactory than expected from the coarse regime separations where $p \sim \delta$ with the layer width δ is basically assumed, although $p \in (0, \infty)$ as discussed. There are, however, several notable differences. There is a significant shift of the $\Im(\hat{\Delta})$ peaks along Q , showing where rotation is stabilizing against error fields and then becomes destabilizing in higher rotation. The peaks are generally greater than what could be inferred by a simple regime connection, as discussed analytically in the Sec. III when $P \gg 1$. In low Q regimes such as HRI, VRii, or Rlii, one can also see the sign flips in $\Im(\hat{\Delta})$. In fact, the regime transitions are so gradual that distinguishing among HRI, VRii, Rlii, SCi, VRi, and Rli would not be necessary. It is also interesting to find the variations and their differences for $\Re(\hat{\Delta})$ in Fig. 4 especially in VRi and I regimes, which can also play a new role in calculating the thresholds of field penetration or tearing mode instabilities.

Another comparison between the analytic and numerical $\hat{\Delta}(Q, P)$ solutions has been made with the conditions more relevant for currently operating high- β tokamak plasmas, with $C=0.4$ and $D=4.0$, with the same setting for $\tau=1$, $Q_e = Q/2$, and $Q_i = -Q/2$. These conditions are relevant to reproduce Fig. 3 in Ref. 40 again with explicit $\hat{\Delta}$ values. Figure 7 shows the $\Im(\hat{\Delta})$ across again with eight different asymptotic regimes but with the second Semi-Collisional (SCii) and second Hall-Resistive (HRii) regimes rather than VRi or Rli. In addition, one can find the subdominant KAW response under the inertial regime when $P \ll 1$ and Q is comparable to several D 's, as qualitatively predicted in the Sec. III. When $P \gg 1$, the KAW response becomes dependent of P , which can be identified as the Visco-Kinetic Alfvén Wave (VKAW) regime. The KAW or VKAW structure is

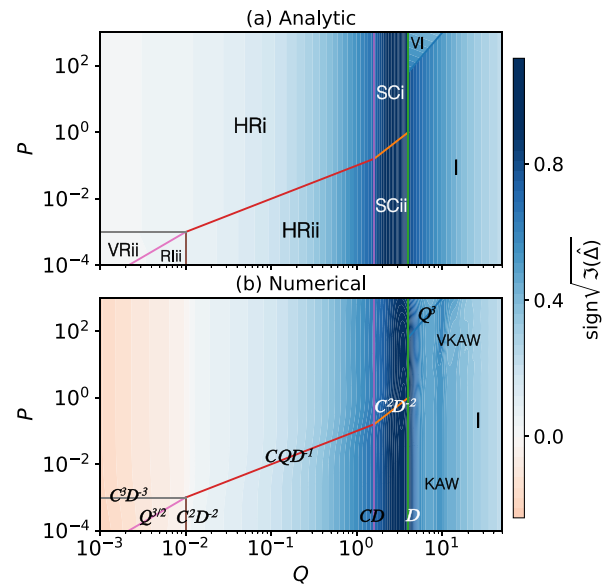


FIG. 7. Comparisons of (a) analytic and (b) numerical solutions for $\Im(\hat{\Delta})$ in the (Q, P) space with $C=0.4$, $D=4.0$, $\tau=1$, and $Q_e = Q/2$ and $Q_i = -Q/2$. The regime identifications are equivalent to Fig. 3 in Ref. 40.

weak but clearly present, as can be more visible in $\Re(\hat{\Delta})$ in Fig. 8 than $\Im(\hat{\Delta})$ in Fig. 7.

These (V)KAW resonances can play a role in the high-temperature tokamak response with $D \gg 1$ in the field penetration or island seeding process when toroidal rotation is also substantial, i.e., when $Q \gg 1$ even if their effects remain subdominant. Figure 9 shows the variations of $1/\Im(\hat{\Delta})$ - approximate $\vec{j} \times \vec{B}$ torque across the layer,

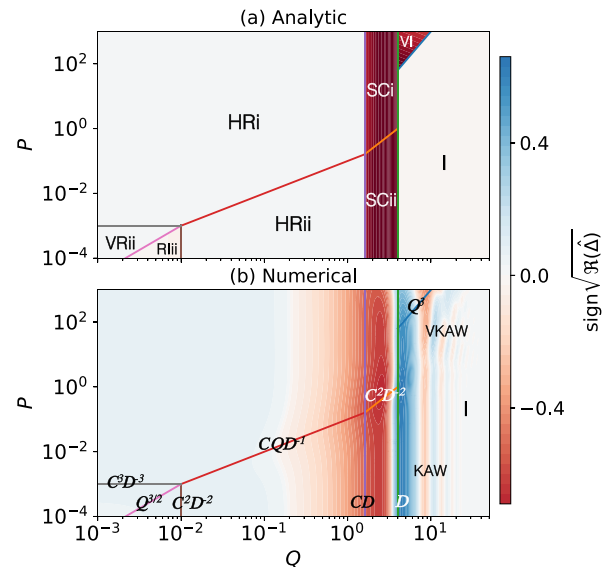


FIG. 8. Comparisons of (a) analytic and (b) numerical solutions for $\Re(\hat{\Delta})$ in the (Q, P) space with $C=0.4$, $D=4.0$, $\tau=1$, and $Q_e = Q/2$ and $Q_i = -Q/2$. The regime identifications are equivalent to Fig. 3 in Ref. 40.

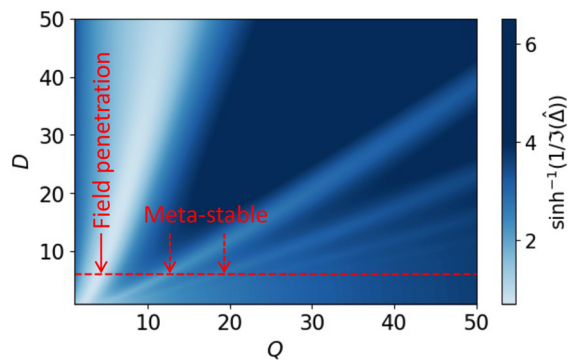


FIG. 9. Comparisons of analytic and numerical solutions for $\Im(\hat{\Delta})$ in the (Q, D) space with $C=0.1$, $P=0.1$, $\tau=1$, and $Q_e = Q/2$ and $Q_i = -Q/2$. Note that the contour has the $\sinh^{-1}$ scale, which is similar to a log scale but can include the sign change. The red line is where $D=6.0$ as an example, to illustrate multiple minima and potential metastable rotational balance as rotation slows down. When Q is sufficiently low to be comparable to Q_e , a field penetration is expected in practice.

in (D, Q) space when $C, P \ll 1$. One can see the local extrema aligning linearly in the (D, Q) space and so clearly indicating the repeated KAW resonances in $\Delta Q \sim D$. The minimum points along with Q then represent the potential metastable rotational equilibrium when the external or internal driving forces are not strong enough. For example, when $D=6.0$ and $Q=20$ initially upon an increasing error field, plasma rotation can reach to the metastable states when $Q \approx 18$ and $Q \approx 12$, and finally $Q \approx 4$ before the full penetration. Note that the full penetration when Q is low enough will be determined mostly likely by the resonance with Q_e (or Q_i if $Q < 0$ or rotation is in the counter- I_p direction), which will be further described in detail by following work in the future.

VI. SUMMARY AND FUTURE WORK

The calculation of the inner-layer Δ remains central to analytically understand the complex responses and dynamics at the resonant layers upon external or internal perturbative driving forces. Accordingly, this paper was devoted to improve the calculations first by following one of the most comprehensive linear layer models in terms of physics but by clarifying the steps as needed. It was shown that only three fields would be needed to reproduce the well-known analytically tractable form, Eq. (20). Further analytic treatments were made to understand the transitions to the inertial regime from the viscous or semi-collisional regime, which illustrated the stronger resonance and subsidiary KAW resonances in resonant responses as well as to verify the numerical solutions. Switching to a numerical approach to avoid the coarse regime separations, the Riccati transformation has been deployed, and its numerical stability has been thoroughly tested. Then, the main results of this paper are to calculate the inner-layer Δ 's across a wide range of parameters, which covers the 11 different regimes identified in the previous literature. The numerical solutions show that the previous asymptotic evaluations are overall satisfactory but resolve further the peaks in the transitions and KAWs across the inertial regimes, sign flips in the opposite low Q regimes as well as the real part of Δ 's. The KAWs were discussed further as they offer predictions or explanations of potential metastable rotational equilibria.

An obvious next step with this formulation is to bring a torque balance model across a single layer and calculate the error field threshold across tokamak parameters. The SLAYER code is fast enough to examine the torque balance and identify the threshold over 10^5 cases in an hour with a single CPU core and, thus, can be used to develop numerical multi-parametric scaling. This can be done immediately but separated from this paper as it can be extensive and somewhat digressive especially due to various possible assumptions for inter-parameter relations. The parametric threshold scaling can be compared with other analytic, numerical, and empirical scaling laws, not only for the error field thresholds against LMs but also for the RMP thresholds for ELM suppression. The goal is to find the applicability and limitation of the linear layer theory, expecting that nonlinear effects can come into play or even dominate depending on cases. The presented scheme can isolate layer physics from global outer-layer responses by providing Δ 's and Δ_{EF} 's in all relevant rational surfaces of realistic toroidal plasmas. This will elucidate relative importances between linear and nonlinear, and between inner-layer and outer-layer response effects.

The linear formulation in this paper can also be extended from three fields to four fields, by restoring the parallel ion motion. It will be difficult to apply asymptotic analytics or the Riccati method on four field equations and, thus, will be necessary to solve them numerically and directly, either in configuration space or Fourier space. The Riccati solutions here will be useful for the purpose of benchmark for the direct numerical solutions by ignoring the parallel ion motion. Interestingly, the Riccati method as well as various asymptotic analytics can still be used on four field equations if the variation of ion parallel motion along the field lines is ignored. It is shown that at least the electron viscosity effect can be included in this scheme. More advanced neoclassical reduced models including important effects such as neoclassical toroidal viscosity (NTV) can also be adopted for linear analysis using similar analytic or numerical schemes to understand its validity in comparison with recent extensive nonlinear analytic treatments.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jong-Kyu Park: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Visualization (lead); Writing – original draft (lead); Writing – review and editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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